IV. THE THREE PLANE APPROACH (TPA) FOR 3D TRUE PROPORTIONAL NAVIGATION

Keeping the essentials of two-dimensional TPN approach explained in Chapter II.B.3.d.(1) in mind, Three Plane Approach (TPA) has been developed and is explained in detail in this chapter. In the TPA; three dimensional (3D) engagement space is projected onto three perpendicular planes: S_{xy}, S_{xz} and S_{yz}. (Figure 11)

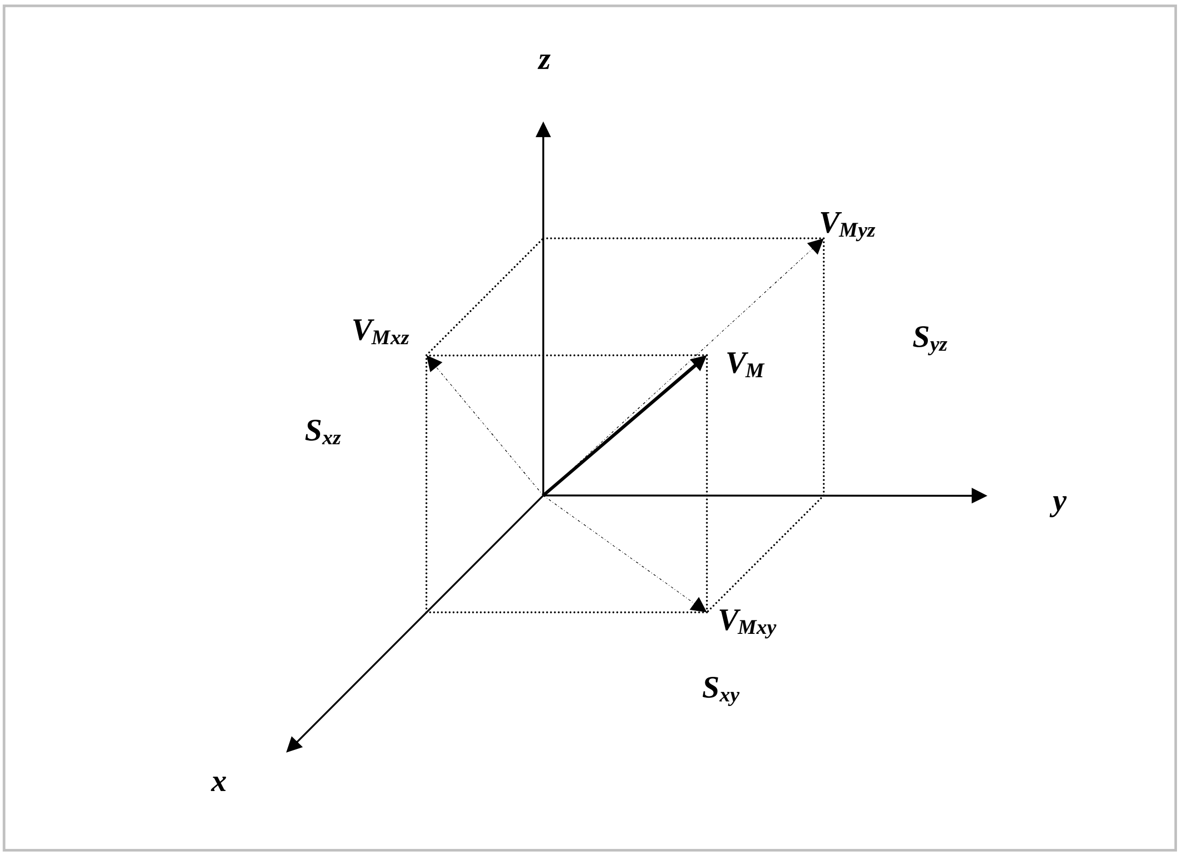
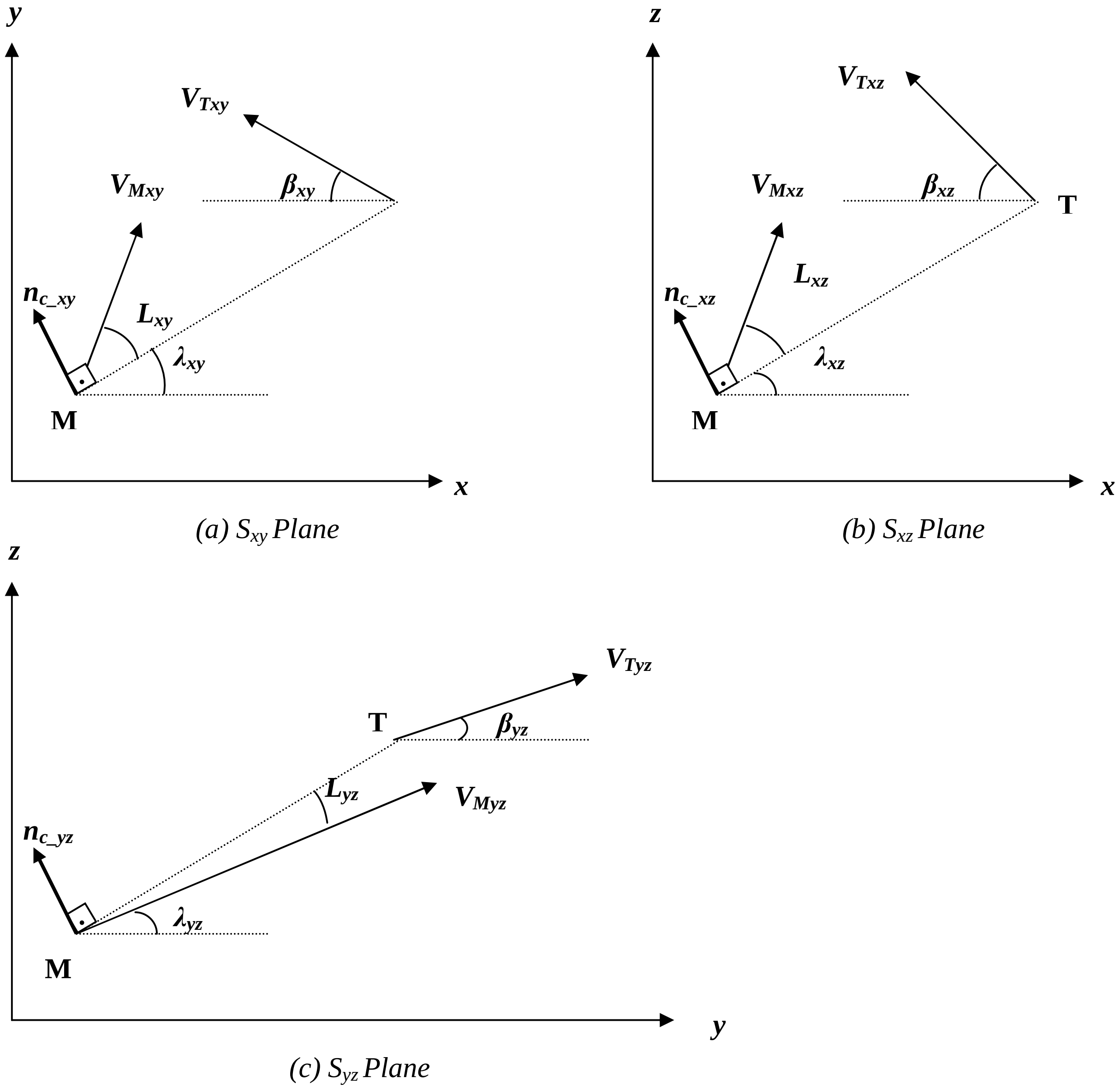


Figure 11 Projections of Missile Velocity Vector on to Three Planes

The projections of missile velocity vector on to the planes are shown in Fig.11. In a pursuit-evasion scenario, there are two main vectors to deal with: target and missile velocity vectors.

It is assumed that the missile and the target are point mass and having the velocity vectors $\boldsymbol{V_T}$, $\boldsymbol{V_M}$ respectively. The projection of such two point masses' relative motion geometry to $\boldsymbol{S_{xy}}$, $\boldsymbol{S_{xz}}$ and $\boldsymbol{S_{yz}}$ planes are shown in Fig. 12 (a),(b),(c).



(a) S_{xy} Plane

(b) S_{xz} Plane

(c) S_{yz} Plane

Figure 12 The Projections of Target's and Missile's Relative Motion onto $\boldsymbol{S_{xy}}$, $\boldsymbol{S_{xz}}$ and $\boldsymbol{S_{yz}}$ Planes

Our approach to solve guidance problem in 3D space is to project 3D geometry onto 3 perpendicular planes such as, $\boldsymbol{S_{xy}}$, $\boldsymbol{S_{xz}}$ and $\boldsymbol{S_{yz}}$; to solve the problem in each plane independently for two-dimensional True Proportional Navigation (TPN) and to produce a 3D solution by combining these 2D solutions.

Considering Fig.11 and Fig.12, the components of 2D solutions and equations to combine those to provide 3D solutions are derived as follows:

The distance between the target and the missile is:

$$R_{TM} = \sqrt{P^2_{TMx} + P^2_{TMy} + P^2_{TMz}} \tag{54}$$

The line of sight (LOS) angles:

$$\lambda_{xy} = \arctan \frac{P_{TMy}}{P_{TMx}} \tag{55 a}$$

$$\lambda_{xz} = \arctan \frac{P_{TMz}}{P_{TMx}} \tag{55 b}$$

$$\lambda_{yz} = \arctan \frac{P_{TMz}}{P_{TMy}} \tag{55 c}$$

Target flight-path angles:

$$\beta_{xy} = \arctan \frac{\Delta P_{Ty}}{\Delta P_{Tx}} \tag{56 a}$$

$$\beta_{xz} = \arctan \frac{\Delta P_{Tz}}{\Delta P_{Tx}} \tag{56 b}$$

$$\beta_{yz} = \arctan \frac{\Delta P_{Tz}}{\Delta P_{Ty}} \tag{56 c}$$

Projection of target velocity vector onto S_{xy}, S_{xz} and S_{yz} planes:

$$V_{Txy} = \sqrt{V^2_{Tx} + V^2_{Ty}} \tag{57 a}$$

$$V_{Txz} = \sqrt{V^2_{Tx} + V^2_{Tz}} \tag{57 b}$$

$$V_{Tyz} = \sqrt{V^2_{Ty} + V^2_{Tz}} \tag{57 c}$$

Missile lead angles, L_{xy}, L_{xz} and L_{yz} for each plane (i.e., S_{xy}, S_{xz}, S_{yz}) have to be computed considering Eq.54-Eq.57 c. In order to find the current leading angles for all possible engagement schemes:

$$L_{xy} = \arcsin \frac{V_{Txy} \, . \, \sin(\beta_{xy} + \lambda_{xy})}{V_M} \tag{58 a}$$

$$L_{xz} = \arcsin \frac{V_{Txz} \, . \, \sin(\beta_{xz} + \lambda_{xz})}{V_M} \tag{58 b}$$

$$L_{yz} = \arcsin \frac{V_{Tyz} \, . \, \sin(\beta_{yz} + \lambda_{yz})}{V_M} \tag{58 c}$$

It can be seen from Eq.5 and Eq.24 that to produce the required acceleration commands for each plane; their closing velocity (V_c) and line of sight (LOS) change rate ($\dot{\lambda}$) values must be computed. From Eq.55 (a)-(c); change rates of LOS angles can be derived as:

$$\dot{\lambda}_{xy} = \frac{P_{TMx} V_{TMy} - P_{TMy} V_{TMx}}{P^2_{TMx} + P^2_{TMy}} \tag{59 a}$$

$$\dot{\lambda}_{xz} = \frac{P_{TMx} V_{TMz} - P_{TMz} V_{TMx}}{P^2_{TMx} + P^2_{TMz}} \tag{59 b}$$

$$\dot{\lambda}_{yz} = \frac{P_{TMy} V_{TMz} - P_{TMz} V_{TMy}}{P^2_{TMy} + P^2_{TMz}} \tag{59 c}$$

To compute the closing velocities(V_{Cxy}, V_{Cxz}, V_{Cyz}) for each plane; P_{TMxy}, P_{TMxz}, P_{TMyz} values must be differentiated just like in Eq.19, thus,

$$V_{Cxy} = -\frac{P_{TMx} . V_{TMx} + P_{TMy} . V_{TMy}}{(P^2_{TMx} + P^2_{TMy})^{1/2}} \tag{60 a}$$

$$V_{Cxz} = -\frac{P_{TMx} . V_{TMx} + P_{TMz} . V_{TMz}}{(P^2_{TMx} + P^2_{TMz})^{1/2}} \tag{60 b}$$

$$V_{Cyz} = -\frac{P_{TMy} . V_{TMy} + P_{TMz} . V_{TMz}}{(P^2_{TMy} + P^2_{TMz})^{1/2}} \tag{60 c}$$

where, relative velocity components are:

$$V_{TMx} = V_{Tx} - V_{Mx} \tag{61 a}$$

$$V_{TMy} = V_{Ty} - V_{My} \tag{61 b}$$

$$V_{TMz} = V_{Tz} - V_{Mz} \tag{61 c}$$

Hence,

$$n_{c\_xy} = N' . V_{Cxy} . \dot{\lambda}_{xy} \tag{62 a}$$

$$n_{c\_xz} = N' . V_{Cxz} . \dot{\lambda}_{xz} \tag{62 b}$$

$$n_{c\_yz} = N' . V_{Cyz} . \dot{\lambda}_{yz} \tag{62 c}$$

acceleration commands for S_{xy}, S_{xz} and S_{yz} planes are derived.

Missile acceleration components (a_{Mx}, a_{My}, a_{Mz}) for x, y and z axis can be computed by combining two components sharing the same axis. Fig.12 indicates that one axe' acceleration component is interacted by 2 planes' acceleration commands. By the help of trigonometric relationships, unified missile acceleration components of axes x, y and z can be founded as below:

$$a_{Mx} = - n_{c\_xy} . \sin \lambda_{xy} - n_{c\_xz} . \sin \lambda_{xz} \tag{63 a}$$

$$a_{My} = n_{c\_xy} . \cos \lambda_{xy} - n_{c\_yz} . \sin \lambda_{yz} \tag{63 b}$$

$$a_{Mz} = n_{c\_xz} . \cos \lambda_{xz} + n_{c\_yz} . \cos \lambda_{yz} \tag{63 c}$$

In the literature, [23, 38, 39, 40] while implementing the acceleration commands as control variables in equations of motion, they are broken into vertical and horizontal components, named as $\boldsymbol{a_{pitch}}$ and $\boldsymbol{a_{yaw}}$.

The vertical acceleration component, $\boldsymbol{a_{pitch}}$ is directed perpendicular to the velocity vector of the missile and upwards, and the horizontal component, $\boldsymbol{a_{yaw}}$ is perpendicular to both the velocity vector and the vertical acceleration component. The suggested vertical and horizontal accelerations generated by Proportional Navigation are defined as:

$$\boldsymbol{a_{pitch} = N' \,.\, V_c \,.\, \dot{\lambda}_{pitch} + g \,.\, \cos} \gamma_M \tag{64}$$

$$\boldsymbol{a_{yaw} = N' \,.\, V_c \,.\, \dot{\lambda}_{yaw}} \tag{65}$$

where, γ_M is flight path angle between the velocity vector and its projection onto the xy plane; $\dot{\lambda}_{pitch}$ and $\dot{\lambda}_{yaw}$ are the line-of-sight rates (LOSR) for relative vertical and horizontal motion respectively. These acceleration commands can not exceed the maximum achievable acceleration limits of the missile imposed by the structural limits.

In our study, vertical and horizontal components of acceleration commands, $\boldsymbol{a_{pitch}}$ and $\boldsymbol{a_{yaw}}$ are computed in a different method. As seen on Fig. 13 and Fig.14, vertical component of missile acceleration command can be derived as:

$$\boldsymbol{a_{pitch} = a_{Mz} \,.\, \cos} \gamma_M + \boldsymbol{g \,.\, \cos} \gamma_M \tag{66}$$

and the lateral component of missile acceleration command:

$$\boldsymbol{a_{yaw} = a_{My} \,.\, \sin} \left(\frac{\pi}{2} - \chi_M \right) - \boldsymbol{a_{Mx} \,.\, \sin \chi_M} \tag{67}$$

These acceleration components, $\boldsymbol{a_{pitch}}$ and $\boldsymbol{a_{yaw}}$, will be used as control variables of the missile in the equations of motion.

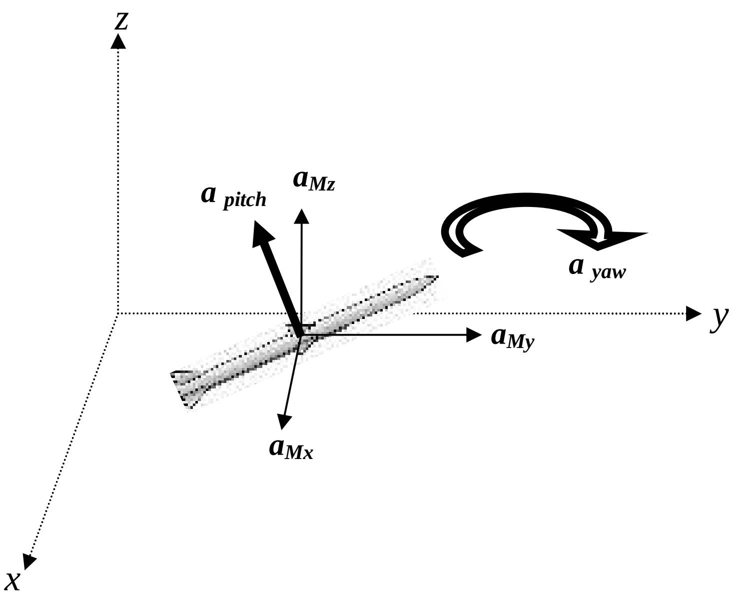


Figure 13 Components of Missile Acceleration Commands in 3D Environment

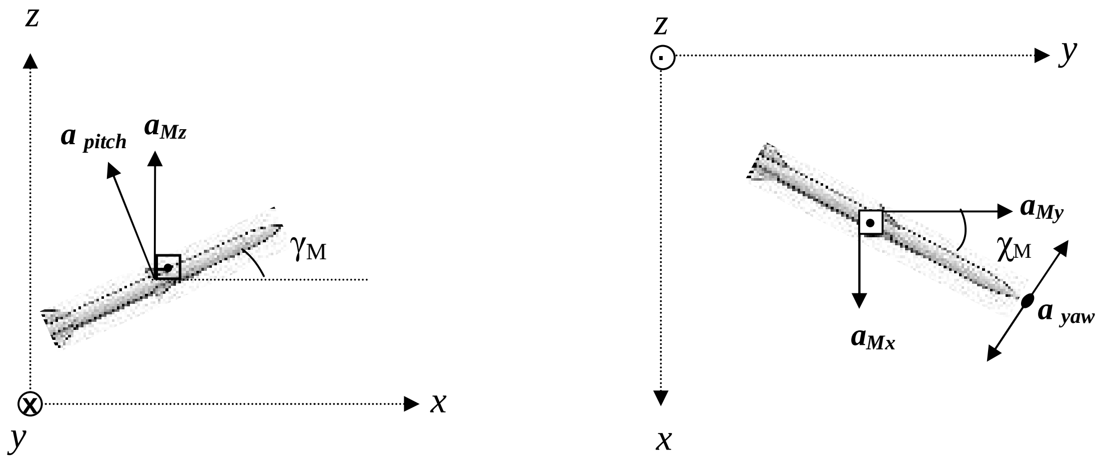


*Figure 14   Projections of Missile Acceleration Commands onto **xz** and **xy** Planes*

In this chapter; Three Plane Approach (TPA) for 3D True Proportional Navigation has been explained.

To summarize TPA, to solve the guidance problem; 3D engagement space geometry is projected onto 3 perpendicular planes such as, $\boldsymbol{S_{xy}}$, $\boldsymbol{S_{xz}}$ and $\boldsymbol{S_{yz}}$. Guidance problem is solved in each plane independently for two-dimensional True Proportional Navigation (TPN); acceleration commands are generated for each plane, and a 3D solution is produced by combining these 2D solutions.

In Chapter VI, TPA is used as missile guidance law and effectiveness of TPA against targets employing evasive maneuvers is examined within aerodynamic limitations.